

**Vidya Vikas Institute of Engineering and Technology,
Mysore
Library and Information Centre**



VTU Question Papers

“COMPUTER SCIENCE & ENGINEERING”

Odd Semester

2022-23

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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CBCS SCHEME

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18CS51

Fifth Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023 Management and Entrepreneurship for IT Industry

Time: 3 hrs.

Max. Marks:100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Management? List the functional areas of management and explain any two in detail. (10 Marks)
- b. Explain in detail levels of management. (06 Marks)
- c. Briefly describe importance of Staffing. (04 Marks)

OR

- 2 a. Discuss the importance of planning Briefly explain the general steps involved in planning. (10 Marks)
- b. Explain any two Management Approaches. (06 Marks)
- c. Write a difference between Formal and Informal organization. (04 Marks)

Module-2

- 3 a. What is Motivation? Explain Maslow's Need Hierarchy theory of Motivation. (10 Marks)
- b. Differentiate between co-ordination and co-operation. (06 Marks)
- c. Write a difference between Manager and a Leader. (04 Marks)

OR

- 4 a. Define Control. Briefly explain the methods of establishing control. (10 Marks)
- b. What are the merits and demerits of Herzberg's two factor theory? (06 Marks)
- c. List the characteristics of Direction. (04 Marks)

Module-3

- 5 a. Define Entrepreneur. Explain the functions of entrepreneur. (10 Marks)
- b. Briefly describe the stages in entrepreneurial process. (06 Marks)
- c. Write the differences between Entrepreneurs and Intrapreneurs. (04 Marks)

OR

- 6 a. Explain different types of entrepreneurs along with classifications. (10 Marks)
- b. What are the barriers of an entrepreneur? (06 Marks)
- c. Write the differences between Entrepreneur and Manager. (04 Marks)

Module-4

- 7 a. Define the meaning of Project. Explain in detail the various ways of Project Identification. (10 Marks)
- b. Briefly describe the contents of Project Report. (06 Marks)
- c. List the different sequential stages for formulation of Project Report. (04 Marks)

OR

- 8** a. Define ERP? Explain the importance and need of a ERP. (10 Marks)
b. Write short notes on functional areas of management of ERP. (06 Marks)
c. List the planning commission guidelines for preparing industrial projects. (04 Marks)

Module-5

- 9** a. Explain the steps involved in establishing Micro and Small enterprise. (10 Marks)
b. Discuss the case study of air Deccan (Captain G.R. Gopinath). (06 Marks)
c. What is Patent? List different types of Patent. (04 Marks)

OR

- 10** a. Explain the following :
i) KIADB
ii) KSSIDC
iii) NSIC
iv) KSFC
v) DIC. (10 Marks)
b. Discuss the case study of Infosys – N.R. Narayana Murthy. (06 Marks)
c. List the advantages of Micro and Small enterprises. (04 Marks)

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18CS52

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Computer Networks and Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the use of cookie files in web applications. (06 Marks)
- b. With a neat diagram, explain how SMTP can be used for transmitting mails from sender to receiver. (08 Marks)
- c. Discuss the working of Bit Torrent for file distribution. (06 Marks)

OR

- 2 a. Differentiate between persistent and non persistent connections in HTTP. (05 Marks)
- b. In brief explain the conditional GET operation. (05 Marks)
- c. Describe the DNS records and messages in detail. (10 Marks)

Module-2

- 3 a. In brief describe UDP segment structure and checksum computation. (06 Marks)
- b. With a neat diagram demonstrate the working of GO-BACK-N protocol. (08 Marks)
- c. Explain TCP flow control in detail. (06 Marks)

OR

- 4 a. With the help of a FSM, describe reliable data transfer in a Lossy channel with bit errors (rdt 3.0). (08 Marks)
- b. Explain the various fields of a TCP segment structure. (05 Marks)
- c. What are the approaches to congestion control? Explain in detail with example. (07 Marks)

Module-3

- 5 a. Explain inter autonomous system routing with Border Gateway protocol. (08 Marks)
- b. Explain various Broadcast Routing algorithms. (08 Marks)
- c. Write a note on IGMP protocol. (04 Marks)

OR

- 6 a. Write the link state algorithm and apply it to the following graph. Assume node 'u' as the source node. (10 Marks)

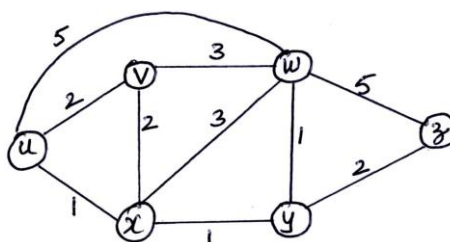


Fig.Q.6(a)

- b. Explain the architecture of a Router. (10 Marks)

Module-4

- 7 a. What are the elements of network security? Discuss the threats to network security. (10 Marks)
- b. Explain RSA algorithm. Using RSA encrypt a message $m = 9$. Assume $p = 3$ and $q = 11$. Find the public key and private key, also show encryption and decryption. (10 Marks)

OR

- 8 a. Explain the working of DES algorithm. (08 Marks)
- b. Discuss the secure Hash Algorithm. (06 Marks)
- c. Write a note on firewalls. (06 Marks)

Module-5

- 9 a. Explain the types of multimedia network applications. (06 Marks)
- b. Briefly explain how DNS redirects a user request to a CDN server. (08 Marks)
- c. With a diagram, explain SIP call establishment. (06 Marks)

OR

- 10 a. What are the properties of video and audio? Explain in detail. (07 Marks)
- b. With a neat diagram, explain streaming stored video over HTTP. (07 Marks)
- c. Explain the Forward Error Correction (FEC) technique for loss anticipation in VoIP application. (06 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Database Management System

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Describe the characteristics of database approach. (08 Marks)
- b. List and explain the criteria for classification of DBMS. (08 Marks)
- c. Write an ER diagram to represent CAR entity type with 2 key attributes Registration and Vehicle ID. (04 Marks)

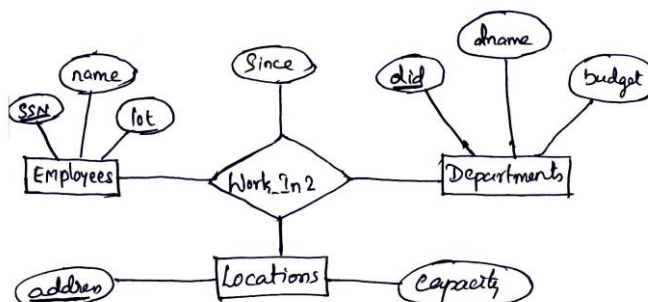
OR

- 2 a. Write an UML class diagram notation for company conceptual schema. (10 Marks)
- b. Define the following terms : i) Data Model ii) Schema iii) Instance (10 Marks)
- iv) Canned transaction v) Data Manipulation Language (DML).

Module-2

- 3 a. Explain the concepts of specialization and Generalization, with the help of VEHICLE superclass. (08 Marks)
- b. Explain the different Relational Model constraints. (06 Marks)
- c. Create a table for the Works_In relationship shown in Fig. Q3(c). (06 Marks)

Fig. Q3(c)



OR

- 4 Considered the COMPANY DATABASE
 EMPLOYEE (Fname , Minit , Lname , Ssn , Bdata , Address , Sex , Salary , Super_Ssn , Dno).
 DEPARTMENT (Dname , Dnumber , Mgr_Ssn , Mgr_Start_data)
 DEPART_LOCATIONS (Dnumber , DLocation)
 PROJECT (Pname , Pnumber , PLocation , Dnum)
 WORKS_ON (ESsn , Pno , Hours)
 DEPENDENT (ESsn , Dependent_name , Sex , Bdate , Relationship).
 Specify the following queries in SQL on the database schema given above.
 - a. For every project located in 'Stafford' , list the project number, the controlling department number and the department managers last name , address and birth date. (06 Marks)
 - b. Retrieve the birth date and address of the employees whose name is 'John B, Smith'. (06 Marks)
 - c. Retrieve the name and address of all employees who work for the 'Research' department. (06 Marks)
 - d. Retrieve the salary of every employee. (02 Marks)

Module-3

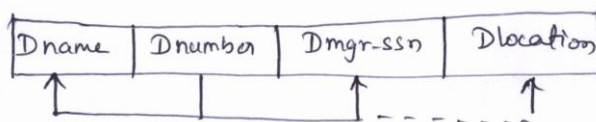
- 5 a. Discuss EXISTS and UNIQUE functions in SQL. Consider the COMPANY database given in Question number 04, write a query to list the name of the Manager who have atleast one dependent. (08 Marks)
- b. With a real World example, explain the following :
- i) JDBC ii) Correlated nested queries iii) Stored Procedures
- iv) Schema change statements in SQL. (12 Marks)

OR

- 6 a. Explain the usage of Aggregate function in SQL. Write an SQL query to find sum of the salaries of all employees, the maximum salary, the minimum salary and the average salary by renaming the columns in a single row table. (10 Marks)
- b. Create an HTML form to collect user id and password fields and it also has to have two buttons one for reset and another for login. (06 Marks)
- c. Write a short note on : i) JavaScript ii) CGI. (04 Marks)

Module-4

- 7 a. Describe the 3 main techniques to achieve first normal form for the relation by taking following examples schema. (04 Marks)



- b. Discuss the Informal guidelines to determine the quality of relations schema design with a suitable example. (10 Marks)
- c. Discuss the Insertion , Deletion and Modification anomalies. Illustrate, why are they considered bad, with an example. (06 Marks)

OR

- 8 a. What do you mean by Normalization? Explain 2NF and BCNF, with a suitable example. (06 Marks)
- b. Consider the universal relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies $F = \{\{A, B\} \rightarrow \{C\}, \{A\} \rightarrow \{D, E\}, \{B\} \rightarrow \{F\}, \{F\} \rightarrow \{G, H\}, \{D\} \rightarrow \{I, J\}\}$.
- i) What is key of R? ii) Decompose R into 2NF and then 3NF relation. (06 Marks)
- c. Write an algorithm to find a minimal cover F for a set of functional dependencies E. (08 Marks)

Module-5

- 9 a. Discuss the ACID properties of database transaction. (08 Marks)
- b. Why concurrency control is needed? Demonstrate with an example. (12 Marks)

OR

- 10 a. Briefly explain 2 phase locking protocols. (05 Marks)
- b. Explain Transaction support in SQL. (05 Marks)
- c. Write a short note on :
- i) Single user and Multiuser system.
- ii) Transaction roll back and Cascading roll back.
- iii) Shadow paging.
- iv) Database backup and recovery from catastrophic failure.
- v) Deadlock prevention protocol. (10 Marks)

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Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Automata Theory and Computability

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define DFSM. Design DFSM
 - i) To accept strings over $\{a, b\}$ such that each block of 5 (length five) consecutive symbols have atleast two a's.
 - ii) To accept $L = \{\omega(ab + ba) \mid \omega \in \{a, b\}^*\}$
 - iii) To accept $L = \{\omega bab \mid \omega \in \{a, b\}^*\}$
- b. Define distinguishable and indistinguishable states. Minimize the following DFSM.

δ	0	1
$\rightarrow A$	B	A
B	A	C
C	D	B
* D	D	A
E	D	F
F	G	E
G	F	G
H	G	D

(10 Marks)

OR

- 2 a. Convert the following NDFSM to DFSM. [Refer Fig.Q2(a)].



Fig.Q2(a)

(08 Marks)

- b. Explain the simulators for Finite State Machine.
- c. Design
 - (i) Mealy Machine that accepts the string that ends either with aa or bb and $\Sigma = \{a, b\}$
 - (ii) Moore Machine that produces 'A', 'B' and 'C' depending on inputs that end with '10', '11' and others respectively.

(06 Marks)

Module-2

- 3 a. Build regular expression from the following FSM. [Refer Fig.Q3(a)].

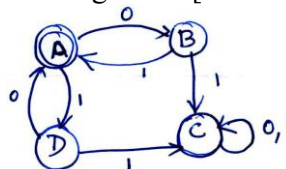


Fig.Q3(a)

(05 Marks)

- b. State and prove pumping Lemma theorem for regular languages. Show that $L = \{a^n b^n \mid n \geq 0\}$ is not Regular.
- c. Show that regular languages are closed under complement and intersection.

(10 Marks)

(05 Marks)

OR

- 4 a. Obtain Regular Expression for the following languages.
- (i) $L = \{ a^n b^m c^p \mid n \leq 4, m \geq 2, p \leq 2 \}$
 - (ii) $L = \{ \omega : |\omega| \bmod 3 = 0 \ \& \ \omega \in \{a, b\}^* \}$
 - (iii) $L = \{ a^n b^m \mid m + n \text{ is even} \}$ (08 Marks)
- b. Prove Kleen's theorem – Any language that can be defined with a regular expression can be accepted by some FSM and so is regular. (08 Marks)
- c. Obtain NDFSM for the following regular expression $(a + b)^* abb$. (04 Marks)

Module-3

- 5 a. Design a PDA for the language
 $L = \{ \omega \omega^R \mid \omega \in (a, b)^* \}$ where ω^R is reverse of ω
 and show the moves made by PDA for the string "aabcbaa" and "abacbbba". (10 Marks)
- b. Define Leftmost derivation, Rightmost derivation and Parse tree. Consider the grammar.
- $$S \rightarrow AbB \quad A \rightarrow aA \mid \epsilon$$
- $$B \rightarrow aB \mid bB \mid \epsilon \quad D \rightarrow a \mid \epsilon$$
- Obtain LMD, RMD and parse tree for the string "aaabab". (10 Marks)

OR

- 6 a. Define CFG and design a CFG for the following language.
- (i) $L = \{ 0^m 1^m 2^n \mid m \geq 1 \text{ and } n \geq 0 \}$
 - (ii) $L = \{ \omega \omega^R \mid \omega \in (a, b)^* \}$
 - (iii) $L = \{ a^n b^m c^k \mid n+2m = k \text{ for } m \geq 0 \text{ and } n \geq 0 \}$ (10 Marks)
- b. Define CNF. Convert the following CFG into CNF.
- $$S \rightarrow ASB \mid \epsilon \quad A \rightarrow aAS \mid a \quad B \rightarrow SbS \mid A \mid bb$$
- (10 Marks)

Module-4

- 7 a. Define TM and design a turing machine for $L = \{ \omega \mid \omega \in (0+1)^* \text{ containing the substring } 001 \}$
 Write transition diagram and show the moves made by the Turing machine for input string 10010. (14 Marks)
- b. Define and explain DTM and NDTM. (06 Marks)

OR

- 8 a. With a neat diagram explain the working of Multitape Turing Machine. (08 Marks)
- b. Design a Turing machine to accept $L = \{ 0^n 1^n \mid n \geq 1 \}$. Show the moves made for the string 0011 and 00111. (12 Marks)

Module-5

- 9 Write short notes on :
- a. Linear Bound Automata (06 Marks)
 - b. Church Turing Thesis (07 Marks)
 - c. Non-Deterministic Turing Machine (07 Marks)

OR

- 10 a. Explain Halting Problem and Post Correspondence problem in Turing Machine. (10 Marks)
- b. Discuss the following :
- i) Decidable and Undecidable Language (05 Marks)
 - ii) Quantum Computers (05 Marks)

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18CS55

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

Application Development using Python

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. List the salient features of python programming language. (04 Marks)
 - b. Explain the math operators in python from highest to lowest precedence with an example for each. Show steps to evaluate $(5 - 1) * (7 + 1) (3 - 1)$ in python. (08 Marks)
 - c. Write a program to find the sum of all odd and even numbers of n elements. Here skip the numbers which are divisible by 3. (08 Marks)

OR

- 2
 - a. Give one example for each of the operation below in python language:
 - i) String concatenation and replication.
 - ii) Read input and display. (10 Marks)
 - b. Explain:
 - i) Def statements with parameters
 - ii) Return values and return statements with an example. (10 Marks)

Module-2

- 3
 - a. Discuss the different ways of traversing a list. Explain each with an example. (10 Marks)
 - b. Write a python program that allows a player to guess a secret number within 6 chances. The code that lets the player enter a guess and checks that guess is right or not by printing appropriate message. List of numbers are taken as an input from the user. (07 Marks)
 - c. Write a program to demonstrate the use of pretty function. (03 Marks)

OR

- 4
 - a. Compare list and dictionary data structures with respect to python language. (04 Marks)
 - b. Write a program in python that counts. The number of occurrences of each letter in a string. Display the results in column fashion. (08 Marks)
 - c. Write the string method syntax in python to perform below operations.
 - i) Removing white space characters from beginning, end or both sides of a string.
 - ii) To right-justify, left-justify and center a string. (08 Marks)

Module-3

- 5
 - a. List out the different character classes. Give representation, regular expression symbols, example and meanings for each character class. (10 Marks)
 - b. Describe the following with suitable python code snippet:
 - i) Greedy and Non Greedy pattern matching.
 - ii) Findall() method of RegeX object. (10 Marks)

OR

- 6 a. Write a python program to rename the filename contains American style dates (MM – DD – YYYY) to European style dates (DD – MM – YYYY) in the working directory. (10 Marks)
- b. What are Assertions in python? Explain with an example. (05 Marks)
- c. Explain the file reading and writing process with suitable python program. (05 Marks)

Module-4

- 7 a. How objects are mutable by nature justify with an example? (04 Marks)
- b. Discuss the methods `isinstance()` and `hasattr()` with suitable example for each. (06 Marks)
- c. What is operator over loading? Write a program to add two point objects by overloading + operator. Also, overload – str – () to display point as an ordered pair. (10 Marks)

OR

- 8 a. Define Inheritance. Explain with an example. (06 Marks)
- b. Briefly discuss the importance of - -init- -() and - - sti - () methods in python. (04 Marks)
- c. Demonstrate the polymorphism to generate histogram to count the number of times each letter appears in word and in sentence. (10 Marks)

Module-5

- 9 a. Analyze the steps involved in downloading and saving web page on to local system along with program. (06 Marks)
- b. List any 4 CSS selectors of bs4 module. Using Beautiful soup passel, retrieve all of the paragraph tags in the web page www.amazon.com. (06 Marks)
- c. How selenium module is useful to deal with web pages. What methods do it uses to simulate mouse clicks and keyboard keys? (08 Marks)

OR

- 10 a. Write a program to read the census data from the excel spreadsheet, count the number of census tracts in each country, count the total population of each country and prints the results.

1	Census tract	State	County	POP2010
9841	06075010500	CA	SanFrancisco	2685
9842	06075010600	CA	SanFrancisco	3894
9843	06075010700	CA	SanFrancisco	5592

- (08 Marks)
- b. How to zip the files and folders. Demonstrate with one example. (04 Marks)
- c. Write a script that will go through every PDF in a folder and encrypt the PDFs using a password provided on the command line. Save each encrypted PDF with an-encrypted pdf suffix added to the original filename. (08 Marks)

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18CS56

Fifth Semester B.E. Degree Examination, Jan./Feb. 2023

UNIX Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Compare internal commands and external commands, files and processes. (06 Marks)
b. Explain all the features of UNIX operating system. (09 Marks)
c. Write the output for the following commands :
i) cal 10 2021
ii) date +%D%T"
iii) type echo
iv) passwd
v) who. (05 Marks)

OR

- 2 a. Explain the different categories of files with examples. (06 Marks)
b. Describe the parent child relationship in UNIX file system and differentiate absolute pathnames with relative path names. (06 Marks)
c. Write the description for the following commands.
i) mkdir college college/ISE college/CSE
ii) mv f₁.C f₂.C f₃.C cprogs
iii) if my pwd is /home/ravi/progs then Cd ../.
iv) ls -l | wc -l
v) cp f₁ f₂ f₃ files
vi) rm -i chap1
vii) cat >> test.txt
viii) rmdir college/ISE (08 Marks)

Module-2

- 3 a. Explain all the options of ls commands with examples. (06 Marks)
b. Consider a file test.txt with default permissions as -rw-r--r--, grant execute permission to owner, write and execute permission to group members and execute permission to others using both relative and absolute approaches. (04 Marks)
c. Write the output for the following commands.
i) cp ??? progs
ii) rm 'chap*'
iii) mv *.[!C][!P][!P] progs
iv) cat *.txt | wc -C
v) cp chap\[0-1\]. (05 Marks)
d. Explain the grep command with all its options. (05 Marks)

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8=50, will be treated as malpractice.

OR

- 4 a. Write a program to read pattern and filename from the user and search the pattern in the given file. (05 Marks)
- b. Write the output for the following commands.
- i) `grep "Anil" std.lst || echo "pattern not found"`
 - ii) `test $x -gt $y`
 - iii) `[-Z $stg]`
 - iv) `[-r $file]`
 - v) `[-n $stg]` (05 Marks)
- c. Explain all the looping statements with syntax. (06 Marks)
- d. Write a shell script to read multiple patterns from the command line and search these patterns in the given file which is also read from command line by using shift command. [Ex. Command line arguments as below `#>script.sh pat1 pat2 pat3 pat4 pat5`]. (04 Marks)

Module-3

- 5 a. Explain the General File API's `open()`, `read()`, `write()`, `lseek()` with their prototype. (10 Marks)
- b. Describe the memory layout of a C program with a diagram and explain memory allocation API's with their prototypes. (10 Marks)

OR

- 6 a. Explain `setjmp` and `longjmp`, `getrlimit` and `setrlimit` function with examples. (10 Marks)
- b. Describe how the process is created by using `fork()` and `vfork()`. List out the properties inherited from the parent when the child process is created? (10 Marks)

Module-4

- 7 a. Explain the implementation of system function using `fork()`, `exec()`, `wait()` API's. (10 Marks)
- b. Define pipes, write a program to send data from parent to child using pipe API and also list its limitations. (10 Marks)

OR

- 8 a. Define semaphores and explain how the IPC is implemented using various semaphore API's. (10 Marks)
- b. Explain the implementation of shared memory IPC mechanism with all its API's and their prototypes. (10 Marks)

Module-5

- 9 a. Define signal and list the actions taken by a process when the signal is raised. Explain the signal API's `signal()`, `sigset()`, `sigaction()`. (10 Marks)
- b. Explain how `kill` API is used for sending a signal to a process and explain the implementation of sleep API using alarm API. (10 Marks)

OR

- 10 a. Define the Daemon process. Explain all the coding rules to be followed while coding a daemon process (10 Marks)
- b. Write a note on interval timer. (05 Marks)
- c. Explain the BSD syslog facility for handling Daemons error messages. (05 Marks)

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Question Paper Version : A

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Which of the following is a possible producer in an ecosystem?
a) Animal b) Plants c) Human beings d) Fish
 2. The largest reservoir of nitrogen in our planet is
a) Oceans b) Biosphere c) Atmosphere d) Rivers
 3. India has the world's largest share of which of the following
a) Manganese b) Mica c) Copper d) Diamond
 4. Identify the non renewable source of energy from the following:
a) Coal b) Fuel cells c) Wind power d) Wave power
 5. Which of the following terminologies is not associated with the vertical structure of forest?
a) Canopy b) Understory c) Forest floor d) First floor
 6. Which of the following is cause of loss of biodiversity?
a) Habitat degradation b) Invasion of non-native species
c) Pollution d) All of these
 7. Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) Cyclone separator
c) Wet collector d) Catalytic converter
 8. When the solid waste consists of large amount of organic matter and if the moisture content is high, which of the following methods of treatment will be ideal?
a) Composting b) Palletizing
c) Incineration d) Recycling

9. Chernobyl Nuclear Disaster occurred in the year
a) 1984 b) 1985 c) 1986 d) 1987
10. The primary cause of acid rain around the world is
a) Carbon dioxide b) Sulphur dioxide
c) Carbon monoxide d) Ozone
11. World Environmental day is held every year on
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16. The most important fuel used by nuclear power plant is
a) U-235 b) U-238 c) U-245 d) U-248
17. Which of the following is a biotic component of ecosystem?
a) Fungi b) Solar light
c) Temperature d) Humidity
18. Abiotic component includes
a) Soil b) Temperature
c) Water d) All of these
19. The word "Environment" is derived from
a) Greek b) French
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a) Geostationary Interact Sector b) Geographical Information System
c) Geotechnical Information Society d) Geothermal Investigation Site
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a) N_2 and H_2S b) CO_2 and N_2
c) Propane and butanes d) Methane and CO_2
83. The Tiger Conservation Project was started in
a) 1973 b) 1975 c) 1981 d) 2000
84. The leader of “Chipko Movement” is
a) Sunderlal Bahuguna b) Medha Patkar
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85. Which of the following is the source of Fly-ash?
a) Vehicular exhaust b) Sewage
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86. The permissible range of pH for drinking water as per the Indian Standard is
a) 6 to 9 b) 6.5 to 8.5
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87. Water logging is a phenomena in which
a) Water patterns are rotated
b) Soil root zone becomes saturated due to over irrigation
c) Erosion of soil
d) Soil degradation
88. Carbon content is higher in
a) Living matter b) Soil
c) Water d) Atmosphere
89. Springs means
a) Surface water b) Atmosphere water
c) Both (a) and (b) d) Ground water

90. Bio-remediation means deliberately introducing micro organisms to break pollutants.
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93. First International Earth Summit was hold at
a) USA
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c) Rio-de-Janerio
d) Johannesburg
94. Which among the following has highest percentage of calorific value?
a) Anthracite
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c) Lignite
d) Bituminous coal
95. Nitrogen fixing bacteria exists in
a) Leaf
b) Stem
c) Roots
d) Flower
96. The two major components of ecosystem are
a) Adiabatic and isotropic
b) Ecologic and climatologic
c) Cyclic and biologic
d) Abiotic and biotic
97. Geothermal energy is a
a) Heat energy
b) Wind energy
c) Current energy
d) Solar energy
98. The average life expectancy around the world is currently.
a) Decreasing
b) Increasing
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99. The universal declaration of Human Rights was proclaimed by the UN in the year.
a) 1946
b) 1947
c) 1948
d) 1949
100. The objective of Integrated Child Development Service (ICDS) are
a) Immunization
b) Health check up and referral services
c) Pre-school non formal education
d) All of these

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Question Paper Version : B

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
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- 90.** Which of the following is absorbed by green plants from the atmosphere?
a) Carbon dioxide b) Water
c) Nutrients d) All of these
- 91.** Which of the following is a possible producer in an ecosystem?
a) Animal b) Plants c) Human beings d) Fish
- 92.** The largest reservoir of nitrogen in our planet is
a) Oceans b) Biosphere c) Atmosphere d) Rivers
- 93.** India has the world's largest share of which of the following
a) Manganese b) Mica c) Copper d) Diamond
- 94.** Identify the non renewable source of energy from the following:
a) Coal b) Fuel cells c) Wind power d) Wave power
- 95.** Which of the following terminologies is not associated with the vertical structure of forest?
a) Canopy b) Understory c) Forest floor d) First floor
- 96.** Which of the following is cause of loss of biodiversity?
a) Habitat degradation b) Invasion of non-native species
c) Pollution d) All of these
- 97.** Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) Cyclone separator
c) Wet collector d) Catalytic converter
- 98.** When the solid waste consists of large amount of organic matter and if the moisture content is high, which of the following methods of treatment will be ideal?
a) Composting b) Palletizing
c) Incineration d) Recycling
- 99.** Chernobyl Nuclear Disaster occurred in the year
a) 1984 b) 1985 c) 1986 d) 1987
- 100.** The primary cause of acid rain around the world is
a) Carbon dioxide b) Sulphur dioxide
c) Carbon monoxide d) Ozone

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Question Paper Version : C

Fifth Semester B.E Degree Examination, Jan./Feb. 2023

Environmental Studies**(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
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Question Paper Version : D**Fifth Semester B.E Degree Examination, Jan./Feb. 2023****Environmental Studies****(COMMON TO ALL BRANCHES)**

Time: 2 hrs.]

[Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **hundred** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

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CBCS SCHEME

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18CS71

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Artificial Intelligence and Machine Learning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. The water jug problem states : You are provided with two jugs, one with 4 gallons of capacity and the other one with 3 gallons of capacity. Neither have any measuring markers on it. How can we get exactly two gallons of water in 4 gallon jug?
 - (i) Write down the production rules for the above problem.
 - (ii) Write any one solution for the above problem. (08 Marks)
- b. Explain Steepest Ascent Hill Climbing technique with an algorithm. Comment on its drawbacks and how to overcome these drawbacks. (12 Marks)

OR

- 2 a. Explain problem reduction with respect to AND-OR graph with suitable example. (07 Marks)
- b. Write AO* algorithm. (07 Marks)
- c. Discuss about constraint satisfaction and solve the below crypt arithmetic problem.

$$\text{CROSS} + \text{ROADS} = \text{DANGER}$$
(06 Marks)

Module-2

- 3 Consider the following sentences :
 1. John likes all kinds of food
 2. Apples are food
 3. Chicken is food
 4. Anything anyone eats and isn't killed is food
 5. Bill eats peanuts and is still alive
 6. Sue eats everything Bill eats.
 - (i) Translate these sentences into formulas in predicate logic. (05 Marks)
 - (ii) Prove that John likes peanuts using backward chaining. (05 Marks)
 - (iii) Convert the formulas of (i) into clause form. (05 Marks)
 - (iv) Prove John likes peanuts using resolution. (05 Marks)

OR

- 4 a. Distinguish forward and backward reasoning with an example. (04 Marks)
- b. Find maximally specific hypothesis for the training instances given below. Also write Find-S algorithm. The concept of this particular problem will be on what days does a person lines to go on walk.

Time	Weather	Temperature	Company	Humidity	Wind	Goes
Morning	Sunny	Warm	Yes	Mild	Strong	Yes
Evening	Rainy	Cold	No	Mild	Normal	No
Morning	Sunny	Moderate	Yes	Normal	Normal	Yes
Evening	Sunny	Cold	Yes	High	Strong	Yes

(08 Marks)

- c. Define version space. Discuss the limitations of finds algorithm over candidate elimination algorithm. (08 Marks)

Module-3

- 5 a. Explain the concept of decision tree learning. Write about attribute selection measure used to build the decision tree using ID3 algorithm. (07 Marks)
- b. How a single perceptron can be used to represent the Boolean functions such as AND, OR. (06 Marks)
- c. Write Gradient Descent algorithm to train a linear unit along with the derivation. (07 Marks)

OR

- 6 a. What do you mean by Gain and entropy? How it is used to build the decision tree. (08 Marks)
- b. Explain back propagation algorithm. Why is it not likely to be trapped in local minima? (08 Marks)
- c. Discuss the perceptron training rule and delta rule that solves the learning problem of perceptron. (04 Marks)

Module-4

- 7 a. Explain Naïve Bayes classifier. (08 Marks)
- b. Explain Bayesian Belief network and conditional independence with example. (08 Marks)
- c. Let us assume dangerous fires are rare (1%) but smoke is fairly common (10%) due to barbecues, and 90% of dangerous fires make smokes. Find the probability of dangerous fire when there is smoke. (04 Marks)

OR

- 8 a. Discuss minimum description length principle in brief. (08 Marks)
- b. Explain brute force MAP learning algorithm. (08 Marks)
- c. Explain EM algorithm. (04 Marks)

Module-5

- 9 a. Explain k-Nearest neighbor learning algorithm. (08 Marks)
- b. Explain Locally weighted regression. (08 Marks)
- c. What is reinforcement learning? (04 Marks)

OR

- 10 a. Distinguish Eager learning vs Lazy learning algorithms. (04 Marks)
- b. Write short notes on Q-learning. (08 Marks)
- c. Discuss about Radial basis function in detail. (08 Marks)

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Seventh Semester B.E. Degree Examination, June/July 2023 Artificial Intelligence and Machine Learning

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Artificial Intelligence? List the applications. (05 Marks)
- b. Solve the 4-gallon water jug problem, writing appropriate production rules. (07 Marks)
- c. Apply A* algorithm and find the most cost effective path from start state A to final state J.

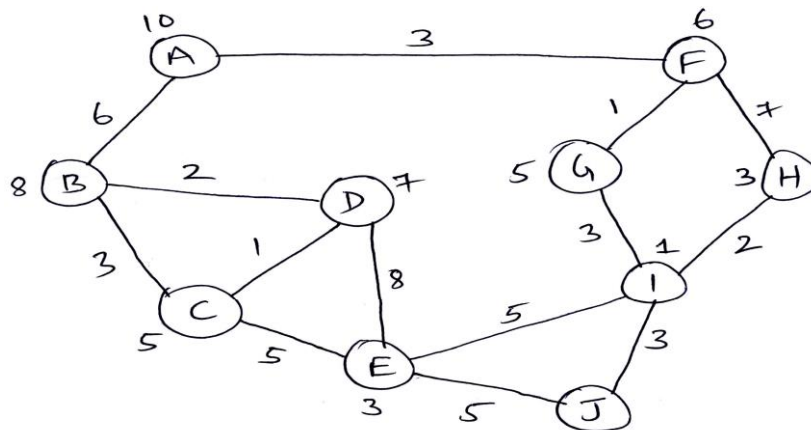


Fig. Q1 (c)

(08 Marks)

OR

- 2 a. What is Heuristic search? Explain the characteristics of heuristic search. (07 Marks)
- b. Provide the solution for water Jug problem by stating the general rules. (07 Marks)
- c. Write the Hill climbing search algorithm of AI. (06 Marks)

Module-2

- 3 a. Discuss the framework of knowledge representation along with any two knowledge representation scheme. (10 Marks)
- b. Explain in brief forward reasoning and backward reasoning. (10 Marks)

OR

- 4 a. Define concept learning. Explain the task of concept learning. (05 Marks)
- b. Write Find-S algorithm. (05 Marks)
- c. Write candidate elimination algorithm. Apply the algorithm for the below dataset to obtain the final version space.

Ex	Citations	Size	Inlibrary	Price	Editions	Buy
1	Some	Small	No	Affordable	One	No
2	Many	Big	No	Expensive	Many	Yes
3	Many	Medium	No	Expensive	Few	Yes
4	Many	Small	No	Affordable	Many	Yes

(10 Marks)

Module-3

- 5 a. Explain the concepts of entropy and information gain in decision tree. (05 Marks)
 b. Apply ID3 algorithm for constructing decision tree for the following training examples. (10 Marks)

Instances	a ₁	a ₂	a ₃	Classification
1	True	Hot	High	No
2	True	Hot	High	No
3	False	Hot	High	Yes
4	False	Cool	Normal	Yes
5	False	Cool	Normal	Yes
6	True	Cool	High	No
7	True	Hot	High	No
8	True	Hot	Normal	Yes
9	False	Cool	Normal	Yes
10	False	Cool	High	Yes

- c. List and explain the appropriate problems for decision tree learning. (05 Marks)

OR

- 6 a. Apply perceptron rule for implement XOR gate by considering the following and compute the final weights.

Inputs : X_1, X_2

Output : y

Initial weights : $W_{11} = W_{21} = 1$

$W_{12} = W_{22} = 1$

$V_1 = V_2 = 1$

Threshold = 1 and learning rate = 1.5

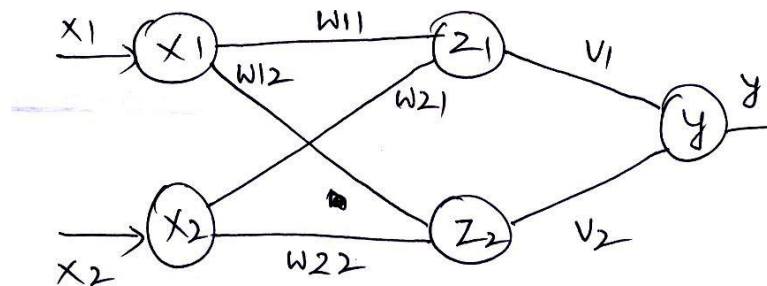


Fig. Q6 (a)

- b. Write Back propagation algorithm. (06 Marks)
 c. Discuss the perceptron training rule and delta rule that solves the learning problem of perceptron. (07 Marks)

Module-4

- 7 a. Explain Bayes theorem and MAP hypothesis with equations. (06 Marks)
 b. Outline Brute force MAP learning algorithm. (08 Marks)
 c. In Orange country, 51% of the adults are males (other 49% are ofcourse females). One adult is randomly selected for a survey involving credit card usage.
 (i) Find the prior probability that the selected person is male.
 (ii) It is later learned that the selected survey subject was Smoking a Cigar. Also, 9.5% of males smoke Cigar, whereas 1.7% of females smoke Cigars. Use this additional information to find the probability that the selected subject is a male. (06 Marks)

OR

- 8 a. Discuss the minimum description length algorithm. (07 Marks)
- b. Apply Naïve Bayes classifier for the below dataset to classify the new instances. (Color = Green, Legs = 2, Height = Tall and Smelly = No)

No.	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

- c. Explain Gibbs algorithm. (06 Marks)

Module-5

- 9 a. Discuss K-Nearest Neighbor learning algorithm. (08 Marks)
- b. Discuss in brief locally weighted linear regression. (06 Marks)
- c. Explain in brief case based reasoning. (06 Marks)

OR

- 10 a. Explain in brief the reinforcement learning technique. (10 Marks)
- b. Discuss the learning tasks and Q learning in the context of reinforcement learning. (10 Marks)

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Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Big data Analytics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Big data. Explain the classification of Big data. (10 Marks)
- b. Define Scalability and its types along with examples. (10 Marks)

OR

- 2 a. Explain the functions of each layer in Big data architecture design with a diagram. (10 Marks)
- b. Define data preprocessing. Explain in brief the needs of preprocessing. (10 Marks)

Module-2

- 3 a. What is HDFS? Highlight the important design features of the HDFS. (10 Marks)
- b. Bring out the concepts of the HDFS block replication with an example. (10 Marks)

OR

- 4 a. Explain Apache sqoop import and export method with neat diagram. (10 Marks)
- b. Demonstrate any six HBase commands with output. (10 Marks)

Module-3

- 5 a. Explain about NOSQL data store and its characteristics. (10 Marks)
- b. Describe the principle of working of the CAP theorem. (10 Marks)

OR

- 6 a. Demonstrate the working of key-value store with an example. (10 Marks)
- b. Describe the features of MangoDB, and its industrial application. (10 Marks)

Module-4

- 7 a. Describe the Map tasks, Reduce tasks and Map Reduce Execution process. (10 Marks)
- b. Describe the Hive architecture and its characteristics. (10 Marks)

OR

- 8 a. Demonstrate the pig architecture for scripts data flow and processing. (10 Marks)
- b. Differentiate between Pig and Map Reduce, give industrial applications for each. (10 Marks)

Module-5

- 9 a. Explain the simple linear regression analysis. (10 Marks)
- b. Demonstrate frequent item set mining and association rule mining. (10 Marks)

OR

- 10 a. With a neat diagram, write the steps in K-means clustering. (10 Marks)
- b. Explain the purpose of web usage analytics and the significance of web graphs. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg, 42+8 = 50, will be treated as malpractice.

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18CS72

Seventh Semester B.E. Degree Examination, June/July 2023

Big Data and Analytics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define big data, and explain its characteristics. (06 Marks)
- b. List and explain different data sources. (04 Marks)
- c. Explain Big data designing architecture. (10 Marks)

OR

- 2 a. Explain the functions of each of the Big Query layers in big data architecture design using Big Query Cloud Service at Google Cloud Platform. (10 Marks)
- b. (i) Explain big data analytics applications. (05 Marks)
- (ii) Write a short note on data storage and analysis. (05 Marks)

Module-2

- 3 a. With a neat diagram, explain the components of HDFS (Hadoop Distribution File System). (10 Marks)
- b. How does the Hadoop MapReduce data flow work for a word count program? Give an example. (10 Marks)

OR

- 4 a. What is APACHE Flume? Describe the feature components and working of apache flume. (10 Marks)
- b. Explain the features and benefits of apache Hive in hadoop. (10 Marks)

Module-3

- 5 a. Discuss NOSQL data architecture pattern, with an example. (10 Marks)
- b. Explain four different ways for handling big data problems. (10 Marks)

OR

- 6 a. (i) Explain different components of Cassandra. (05 Marks)
- (ii) Explain different data types built into Cassandra. (05 Marks)
- b. (i) Describe different CQL commands and their functionalities. (05 Marks)
- (ii) Write a short note on NOSQL to Manage Big Data. (05 Marks)

Module-4

- 7 a. Describe the significance of apache pig in hadoop. (10 Marks)
- b. With a neat diagram, explain MapReduce Programming model. How does MapReduce enables query processing quickly in Big Data Problems? (10 Marks)

OR

- 8 a. Explain Hive architecture with a neat diagram. (10 Marks)
- b. (i) Differentiate between Pig and MapReduce. (05 Marks)
- (ii) Write a short note on Pig architecture design layers. (05 Marks)

Module-5

- 9 a. What are outliers? Describe the reasons for the presence of outliers in a relationship. (10 Marks)
- b. How can a university student's GPA be predicted from his/her – high school percentage (HSP) of marks? (Assume linear regression) Plot a graph for the same. (10 Marks)

OR

- 10 a. Discuss the different phases of text mining process. (10 Marks)
- b. Write a short note on text mining and web mining. (05 Marks)
- c. Discuss three phases for web usage mining. (05 Marks)

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18CS734

Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

User Interface Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define User Interface. Discuss the benefit of good design. (10 Marks)
b. Discuss the characteristics of the Graphical User Interface. (10 Marks)

OR

- 2 a. Compare GUI versus Web Page Design with respect to Devices, Data, Presentation Elements, Navigation, User tasks. (10 Marks)
b. Discuss the following general principles of user interface
i) Aesthetically pleasing ii) Compatibility iii) Consistency
iv) Control v) Simplicity. (10 Marks)

Module-2

- 3 a. List and explain common usability problems in graphical systems. (10 Marks)
b. Discuss any five Human characteristics in interface design. (10 Marks)

OR

- 4 a. Explain Indirect methods of requirement determination in Business Function. (10 Marks)
b. Discuss briefly the guidelines for designing conceptual models. (10 Marks)

Module-3

- 5 a. List and explain different structures of Menus with suitable diagrams. (12 Marks)
b. Discuss Functions and content of Menus. (08 Marks)

OR

- 6 a. Explain the purpose, advantages, disadvantages, guidelines to be followed in designing following menu choices. i) Mark Toggles ii) Toggled Menu Items. (08 Marks)
b. Discuss in detail the following Graphical menus
i) Pull – Down Menu ii) Cascading Menus iii) Popup Menus. (12 Marks)

Module-4

- 7 a. List and discuss in different ways, windows are useful. (12 Marks)
b. Explain filed windows, overlapping windows, and cascading windows their advantages and disadvantages. (08 Marks)

OR

- 8 a. Explain model and modeless and cascading and unfolding windows. (08 Marks)
b. Explain the following with respect to windows
i) frame ii) Window sizing Buttons iii) Scroll bars iv) Split box. (08 Marks)
c. Describe Joystick with advantages and disadvantages. (04 Marks)

Module-5

- 9 a. Explain different command button guide lines. (12 Marks)
b. Describe check boxes, list boxes, palettes with advantages and disadvantages. (08 Marks)

OR

- 10 a. List common presentation controls and discuss any four of them. (08 Marks)
b. Explain the following kinds of tests
i) Cognitive walk through ii) Think-Aloud Evaluations iii) Usability Test. (12 Marks)

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18CS734

Seventh Semester B.E. Degree Examination, June/July 2023

User Interface Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define User Interface. Explain its two essential components. (06 Marks)
- b. List and explain the characteristics of GUI. (06 Marks)
- c. Define object in graphical system. Differentiate between application and data orientation. (08 Marks)

OR

- 2 a. Explain the importance and benefits of good design. (07 Marks)
- b. Explain the characteristics of intranet and internet. Also write the differences between them. (08 Marks)
- c. Write any five differences between GUI and Web page design. (05 Marks)

Module-2

- 3 a. List and explain the pitfalls in the development path of the design process. (08 Marks)
- b. Describe the characteristics of Human Interface Design. (12 Marks)

OR

- 4 a. Explain common usability problem in Web Based Systems. (10 Marks)
- b. Explain techniques for determining the user requirement using indirect method. (10 Marks)

Module-3

- 5 a. Explain and illustrate the structure of menus. (10 Marks)
- b. Describe the components of web navigation systems with illustration. (10 Marks)

OR

- 6 a. Explain various guidelines to be followed in phrasing of menu. (10 Marks)
- b. List all kinds of graphical menus and explain them. (10 Marks)

Module-4

- 7 a. What are the importances of windows? Explain. (12 Marks)
- b. Explain any two window management schemes in detail. (08 Marks)

OR

- 8 a. Explain the advantages and disadvantages of frames in web. (10 Marks)
- b. Explain the characteristics of the devices touch screen and keyboard. (10 Marks)

Module-5

- 9 a. What are operable controls? Explain the usage of buttons along with their advantages and disadvantages. (10 Marks)
- b. Explain the purpose of prototypes. Discuss any two kinds of prototypes with their importance to the system developers. (10 Marks)

OR

- 10 a. Distinguish between the following :
- (i) Radio buttons and Check boxes.
 - (ii) Tool tips and Ballon tips. (10 Marks)
- b. Explain Cognitive Walkthroughs and Think Aloud evaluation method to perform test in user interface design. (10 Marks)

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Seventh Semester B.E. Degree Examination, Jan./Feb. 2023

Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

1. a. With a neat block diagram, explain the fundamental steps in image processing. (10 Marks)
- b. Explain the concept of sampling and quantization of an image. (05 Marks)
- c. Explain any four fields that use digital image processing. (05 Marks)

OR

2. a. Define 4-, 8-, m-adjacency with appropriate examples. (04 Marks)
- b. Explain the various distance measures used with suitable examples. (06 Marks)
- c. Consider the two image subsets S1 and S2 as shown in Fig. Q2 (c) for $V = \{1\}$. Determine whether these two subsets are (i) 4-adjacent (ii) 8-adjacent (iii) M-adjacent

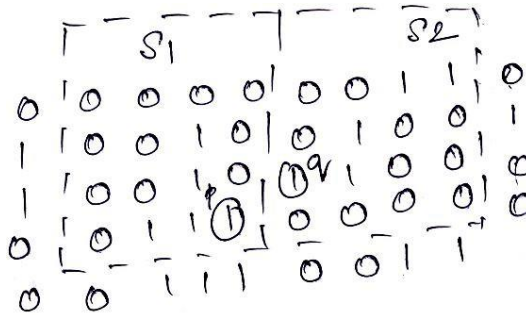


Fig. Q2 (c)

(10 Marks)

Module-2

3. a. Explain in detail the basic gray level transformations used in image enhancement. (10 Marks)
- b. Define image enhancement. Explain how arithmetic operators are helpful in image enhancement. (10 Marks)

OR

4. a. Define histogram and normalized histogram. Discuss histogram equalization for contrast enhancement. (10 Marks)
- b. Explain image smoothing in spatial domain. (03 Marks)
- c. For a given 4×4 image having gray scales between $[0, 9]$ perform histogram equalization and draw the histogram equalization and draw the histogram of image before and after equalization. 4×4 image is shown in Fig. Q4 (c).

2	3	3	2
4	2	4	3
3	2	3	5
2	4	2	4

Fig. Q4 (c)

(07 Marks)

Module-3

- 5 a. Explain the derivation of DFT from the continuous transform of a sampled function. (10 Marks)
b. Explain homomorphic filter with procedure for applying the same. (10 Marks)

OR

- 6 a. Write the properties of Discrete Fourier transforms. (10 Marks)
b. Discuss Butterworth Low Pass filters and Gaussian Low Pass frequency domain filters. (10 Marks)

Module-4

- 7 a. Explain various Gradient operators used for edge detection. (10 Marks)
b. Explain Global Thresholding using Otsu's method. (10 Marks)

OR

- 8 a. Explain the various approaches used for edge linking. (10 Marks)
b. Explain the basic of intensity thresholding in image segmentation. (10 Marks)

Module-5

- 9 a. Explain in detail the principal types of data redundancies. (10 Marks)
b. Explain Huffman coding technique with an example. (10 Marks)

OR

- 10 a. Explain Lossy and Lossless compression. (10 Marks)
b. Explain Arithmetic coding technique with an example. (10 Marks)

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18CS741

Seventh Semester B.E. Degree Examination, June/July 2023 Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define an image. With block diagram, explain the fundamental steps in digital image processing. (12 Marks)
- b. Explain 4-connectivity, 8-connectivity and m-connectivity with suitable example. (08 Marks)

OR

- 2 a. With necessary diagrams, discuss image digitization process. (10 Marks)
- b. Discuss any four applications of digital image processing. (10 Marks)

Module-2

- 3 a. Define histogram equalization. Develop an algorithm to enhance image quality using this method. (12 Marks)
- b. Discuss any three gray level transformation techniques with suitable graph plots. (08 Marks)

OR

- 4 a. Explain the importance of Sobel and Prewitt operators in image processing. (08 Marks)
- b. Discuss Laplacian and Log Edge detectors with suitable mathematical model. (12 Marks)

Module-3

- 5 a. List the properties of DFT. Explain Discrete Fourier Transform Process along with its computational complexity. (12 Marks)
- b. Discuss steps in frequency domain based filtering techniques. (08 Marks)

OR

- 6 a. Discuss the filters used to smooth an image in frequency domain. (10 Marks)
- b. Explain the significance of homomorphic filters with a block diagram. (10 Marks)

Module-4

- 7 a. Define image segmentation. What are the properties of segmentation? (10 Marks)
- b. Explain 'Region Growing' and 'Split and Merge' approach used in image segmentation. (10 Marks)

OR

- 8 a. Explain how Hough Transform helps in extracting line segments from an image. (10 Marks)
- b. Define thresholding. Explain Global thresholding technique with an example. (10 Marks)

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and/or equations written eg. 42+8=50, will be treated as malpractice.

Module-5

- 9 a. Define image compression. Explain general image compression model with a block diagram. (10 Marks)
- b. Develop an algorithm that encodes data using Huffman coding technique. Illustrate Huffman coding for the following data:

Data	S ₁	S ₂	S ₃	S ₄	S ₅	S ₆
Probability	0.1	0.4	0.06	0.1	0.04	0.3

(10 Marks)

OR

- 10 a. Discuss LZW encoding and decoding technique with an example. (12 Marks)
- b. With a block, discuss transform coding technique. (08 Marks)

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Seventh Semester B.E. Degree Examination, June/July 2023

Python Application Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List the rules to declare variable in python and salient features of python. (06 Marks)
- b. Write a python program to find biggest of three numbers accepted from the user. (06 Marks)
- c. Explain concept of type conversion function and math functions in python with examples. (08 Marks)

OR

- 2 a. Explain break and continue statement with example in python. (06 Marks)
- b. How python handles the exceptions? Explain with an example program. (06 Marks)
- c. What are user defined functions? Write a function compute grade that takes a score as its parameter and returns a grade as string.

Score	Grade
> = 0.9	A
> = 0.8	B
> = 0.7	C
> = 0.6	D
> = 0.6	F

(08 Marks)

Module-2

- 3 a. Discuss string handling methods in python with examples. (08 Marks)
- b. Write a python program to demonstrate counting, summing and average of elements using loops. (06 Marks)
- c. Write a python program to check whether given string is palindrome or not. (06 Marks)

OR

- 4 a. Explain for loop and while loop and write a program to generate Fibonacci series up to the given limit. (08 Marks)
- b. Write a python program to count the number of occurrences of a word in a given file. (06 Marks)
- c. Write a program to read through a file and look for lines of the form.
X-DSPAM – confidence : 0.8475
When you encounter line that starts with “X – DSPAM confidence” pull apart the line to extract the floating points numbers on the line, count there line and then compute the total of the spam confidence. (06 Marks)

Module-3

- 5 a. What are lists? Describe list methods in python. Write a program to find biggest element in the list. (08 Marks)
- b. How tuples are created in python? Explain different ways of accruing and creating them. (06 Marks)
- c. Write python program to search for lines that start with ‘F’ followed by 2 characters, followed by ‘m’ and print there lines. (06 Marks)

OR

- 6 a. What is dictionary? Write a python program to count occurrence of words in a file and print the count. (08 Marks)
- b. Explain extracting data using regular expressions implement python program to find for lines having '@' symbol between non blank characters in a file. (08 Marks)
- c. Compare and contrast lists and tuples. (04 Marks)

Module-4

- 7 a. Explain init and str method with an example python program. (10 Marks)
- b. What is Operator overloading? Write python code to overload "+", "-" and "*" operators by providing the methods __add__, __sub__ and __mul__. (06 Marks)
- c. Define class and object. Explain with example. (04 Marks)

OR

- 8 a. What is a pure function? Write a python program to find duration of event if start and end time is given by defining class TIME. (10 Marks)
- b. Write a python program that uses date time module within a class takes birthday as input and points the age and the no of days, hours, minutes and seconds until next birthday. (10 Marks)

Module-5

- 9 a. Demonstrate with the help of python construct
- How to retrieve an image over HTTP
 - How to retrieve web pages with Urllib. (10 Marks)
- b. Write a note on Google Geocoding Web service. Using python supported libraries demonstrate with snippet code. (10 Marks)

OR

- 10 a. Write a python program for creating employee database, inserting records and selecting employers working in the company. (10 Marks)
- b. Describe creation of database table using database cursor architecture. (10 Marks)

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Third Semester B.E. Degree Examination, Jan./Feb. 2023

Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. What is linear array? Discuss the representation of linear array in memory. (06 Marks)
 - b. Differentiate between static and dynamic memory allocations. Discuss four dynamic memory allocation functions. (06 Marks)
 - c. Write a menu driven program in C for the following array operations:
 - (i) Inserting an element (ELEM) at a given valid position.
 - (ii) Deleting an element at a given valid position.
 - (iii) Display of array elements.
 - (iv) Exit
 Support the program with functions for each of the above operations. (08 Marks)

OR

- 2
 - a. Give Abstract Data Type (ADT) for arrays. How array can be declared and initialized? (06 Marks)
 - b. With suitable example, discuss self-referential structures. (06 Marks)
 - c. Define Sparse matrix. How to represent a Sparse matrix? Write an algorithm/function to transpose a given Sparse matrix. (08 Marks)

Module-2

- 3
 - a. Define Stack. Discuss how to represent stack using dynamic arrays. (06 Marks)
 - b. Write a menu driven C program for the following operations on STACK of integers:
 - (i) Push an element on to stack
 - (ii) Pop an element from the stack
 - (iii) Display the content of stack
 - (iv) Exit
 Show the overflow and underflow conditions. (06 Marks)
 - c. What are the disadvantages of ordinary queue? Discuss the implementation of circular queue using arrays. (08 Marks)

OR

- 4
 - a. What is Recursion? Write recursive function to solve Towers of Hanoi problem. (06 Marks)
 - b. Discuss the following:
 - (i) Double Ended Queue
 - (ii) Priority Queue(06 Marks)
 - c. Write an algorithm to convert infix expression to postfix expression. Show the content of stack to convert the following infix expression:

$$A * (B + D) / E - F * (G + H / K)$$
(08 Marks)

- 5 a. Write a C function to concatenate two singly linked list. (06 Marks)
b. Give the structure definition for singly linked list. Write a C function to:
(i) Insert an element at the end
(ii) Delete a node at the beginning (08 Marks)
c. Discuss how to read a polynomial consisting of 'n' terms implemented using linked list. (06 Marks)

OR

- 6 a. Write a function to delete a node whose information field is specified in singly linked list. (06 Marks)
b. What is circular doubly linked list? Write a C function to perform the following operations on circular doubly linked list:
(i) Insert a node at the beginning
(ii) Delete a node from the list (08 Marks)
c. Discuss how to implement stacks and queues using linked list. (06 Marks)

Module-4

- 7 a. Define binary tree. List and discuss any two properties of binary tree. (06 Marks)
b. Write a function to perform the following operations on Binary Search Tree (BST):
(i) Deletion from a BST
(ii) Inserting an element into a BST (08 Marks)
c. Define Threaded Binary Tree. Discuss In-threaded binary tree. (06 Marks)

OR

- 8 a. Discuss how binary tree are represented using (i) Array (ii) Linked list (06 Marks)
b. Discuss inorder, preorder, postorder and level order traversal with suitable recursive function for each. (08 Marks)
c. Write a C function to evaluate an expression using expression tree. (06 Marks)

Module-5

- 9 a. Design a C program for the following operation on Graph (G) of cities:
(i) Create a graph of N cities using adjacency matrix
(ii) Print all the nodes reachable from a given starting node in a digraph using BFS/DFS method (10 Marks)
b. Discuss AVL tree with an example. Write a function for insertion into an AVL tree. (10 Marks)

OR

- 10 a. Define hashing. What are the two criteria, a good hash function should satisfy? Discuss open addressing and chaining method with an example. (10 Marks)
b. Define Red-Black tree, Splay tree and B tree. Discuss the method to insert an element into Red-Black tree. (10 Marks)

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Third Semester B.E. Degree Examination, Jan./Feb. 2023
Analog and Digital Electronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Derive an expression for collector current and collector emitter voltage of voltage divider bias circuit (accurate analysis). (08 Marks)
- b. Explain relaxation oscillator. (06 Marks)
- c. Sketch and explain the working of Peak detector. (06 Marks)

OR

- 2 a. Explain R-2R ladder type DAC with a neat diagram. (06 Marks)
- b. List the advantages of active filters over passive filters. (06 Marks)
- c. For the circuit shown in Fig. Q2 (c) below find the value of R_1 and R_2 if supply voltages are +12 and -12 V. Assume hysteresis with -6 V.

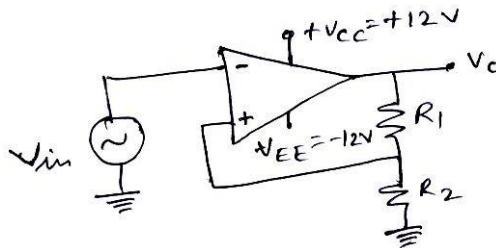


Fig. Q2 (c)

(08 Marks)

Module-2

- 3 a. Find all the prime implicants of the function,
 $f(a, b, c, d) = \Pi(0, 2, 3, 4, 5, 12, 13) + \Pi d(8, 10)$
 using the Quine-McCluskey method. (10 Marks)
- b. Plot the Karnaugh maps and find all the minimal sums and minimal products of the following Boolean functions.
 (i) $f(a, b, c) = \sum(2, 4, 5, 6, 7)$
 (ii) $f(a, b, c) = \Pi(1, 4, 5, 6)$ (10 Marks)

OR

- 4 a. With an example, explain Petrik's method. (06 Marks)
- b. For the given Boolean function, determine a minimal sum and a minimal product using MEV techniques using a, b and c as the map variables.
 $f = \sum(3, 4, 5, 7, 8, 11, 12, 13, 15)$ (08 Marks)
- c. Explain Entered variable map method. (06 Marks)

Module-3

- USN** a. Explain the importance of three-state buffer. **21CS33**
(06 Marks)
b. With a neat diagram, explain 3 to 8 line decoder. (06 Marks)
c. What is a multiplexer? Write the logic diagram for 8 : 1 multiplexer using 4 input AND and OR gates. (08 Marks)

OR

- 6** a. Discuss different types of hazards in combinational circuits. (08 Marks)
b. Distinguish between combinational and sequential circuit. (06 Marks)
c. Write a note on PLA and PAL. (06 Marks)

Module-4

- 7** a. Explain the working of JK master slave flip-flop with a sketch, truth table and symbol. (06 Marks)
b. What is D flip flop? Illustrate the operation of the clear and preset inputs in D-flip-flop with timing diagram. (08 Marks)
c. What is VHDL? Show how to model the 4 to 1 multiplexer using a VHDL conditional assignment statement. (06 Marks)

OR

- 8** a. What is T-flip-flop? Show how to convert D-flip flop into T-flip-flop. (08 Marks)
b. What are the three different models for writing a module body in VHDL? Give example for any one model. (06 Marks)
c. Explain with a neat diagram, VHDL program structure. (06 Marks)

Module-5

- 9** a. With a neat diagram, explain 4-bit parallel adder with accumulator. (10 Marks)
b. Define counter. Design mod-5 counter using J-K flip flop. (10 Marks)

OR

- 10** a. With neat diagram, explain 4 bit SISO register. (08 Marks)
b. Mention the Application of shift registers. (05 Marks)
c. Explain the working of a 3 bit shift register. (07 Marks)

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Third Semester B.E. Degree Examination, Jan./Feb. 2023 Computer Organization and Architecture

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With the help of a neat block diagram discuss the basic operational concept of a computer. (08 Marks)
- b. Write a program to evaluate the arithmetic statement $Y = (A + B) * (C + D)$ using three address, two address, one address and zero address instruction. (08 Marks)
- c. Write the basic performance equation indicate the role of each parameter in the equation. (04 Marks)

OR

- 2 a. Define Addressing Mode. Explain the various addressing mode. (10 Marks)
- b. With proper example explain Big - Endian and Little - Endian of byte addressing. (06 Marks)
- c. What is performance measurement? Explain the overall SPEC rating of a computer. (04 Marks)

Module-2

- 3 a. With respect to handling interrupts from multiple devices explain:
(i) Interrupt nesting (10 Marks) (ii) Dairy chain method.
- b. What is Bus arbitration? Explain centralized and distributed arbitration method with neat diagrams. (10 Marks)

OR

- 4 a. Illustrate a program that reads one line from keyboard, stored it in memory buffer and echoes if back to display in I/O interfaces. (10 Marks)
- b. Discuss with a neat circuit diagram, the general 8 bit parallel interface circuit. (10 Marks)

Module-3

- 5 a. Explain the internal organization of 16-megabit DRAM chip configured as $2M \times 8$. (08 Marks)
- b. With a neat figure illustrate the structure of synchronous DRAM (SDRAM). (08 Marks)
- c. Discuss about any two types of Read Only Memory (ROM). (04 Marks)

OR

- 6 a. State the importance of cache memory and describe the different types of cache mapping techniques with diagram. (12 Marks)
- b. With relevant figure explain organization of $(1k \times 1)$ memory chip. (08 Marks)

Module-4

- 7 a. With the help of logic diagram explain 4-bit carry look adder and its operation. (10 Marks)
- b. Illustrate the hardware arrangement for sequential multiplication with an example. (10 Marks)

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OR

- 8** a. Draw the single bus architecture and explain the control sequence for execution of instruction ADD (R3), R1. (10 Marks)
b. With neat sketches, explain the detailed organization of hardwired control unit. (10 Marks)

Module-5

- 9** a. With a suitable example explain the concept of pipeline processing. (10 Marks)
b. Draw and explain pipeline for floating point addition and subtraction. (10 Marks)

OR

- 10** a. With the help of flowchart and timing diagram explain four segment instruction pipeline. (10 Marks)
b. Explain the organization of SIMD array processor with an appropriate diagram. (10 Marks)

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Question Paper Version : A

Third Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Fill in the blank by translating the given English word to Kannada :

1. You : _____
a) neenu b) Naanu c) yaaru d) Ivaru
2. His : _____
a) NeeVu b) avaru c) aVaLu d) avana
3. What : _____
a) idu b) adu c) yenu d) yelli
4. Friend : _____
a) Gumpu b) Maava c) Sneheta d) GurugaLu
5. Drama : _____
a) naataka b) Kate c) haaDu d) Yaavadu alla
6. Choose the Mathematical + Symbol in Kannada
a) Kale b) Koodu c) Jodisu d) Bhagisu
7. Country in Kannada will be called as
a) D'weshu b) Vesha c) Desha d) Vishesha

Note : Substitute the word from the following each sentence in appropriate places :

8. idu : idakke then adu _____
a) adakke b) idakke c) Horakke d) Horakkee
9. UURu : Urige then avaru _____
a) ivarige b) avarige c) Nimage d) Nawage

10. Mane : maneyalli then Pustaka
a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
11. Raja : Rajanaddu, then Raani _____
a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Write the English word for the given Kannada word :

12. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
13. pramaaNu : _____
a) promise b) very good c) purpse d) Thankyou
14. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
15. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
16. biLi : _____
a) red b) Black c) White d) Green
17. doDDa : _____
a) Big b) Small c) fat d) thin
18. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
19. gurugaLu
a) Former b) Teacher c) Father d) Uncle
20. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast
21. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give
22. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
23. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
24. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
25. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
26. huTTu : _____
a) Good b) die c) Birth d) bed

27. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
28. daKShiNa : _____
a) West b) East c) South d) North
29. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
30. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
31. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
32. granthalaya : _____
a) House b) office c) college d) Library
33. baNNa : _____
a) White b) Banana c) Colour d) Red

Note : Translate the English question to Kannada :

34. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
35. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
36. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla
37. When are you going to College?
a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
38. Had your breakfast?
a) YaaVagu Tindi? b) TinDi ayitaa?
c) oota ayta? d) TinDi yestu beku?

Note : Translate the Question Kannada to English :

39. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?
40. Ee pustaKa yaaraduu?
a) Is that book yours? b) whose book is this?
c) This book belongs to me. d) none

41. ethaKKe hogidde?
 a) Why did you go? b) Where did you go?
 c) When you are going? d) None of these
42. nimma Kalejinaa hesarenu?
 a) What is your college name? b) Where is your college
 c) How old is your college? d) Which college are you studying?
43. naanu yenu MaDabhahudu?
 a) What did you do? b) What should we do?
 c) What can I do? d) None
44. eegale baa
 a) come immediatly b) Go immediatly c) come Tomorrow d) None
45. dayamaDi niiVu oLagaDe banni
 a) When you come b) Come inside
 c) Don't come hear d) you please come inside
46. Kallegige raje KoTTi ddare
 a) Holiday is given to college b) Don't come to college
 c) Today college is there d) None
47. Raamanu nanaginta doDDavanu
 a) Rama is older than me b) Rama is elder than me
 c) Rama is my friend d) None
48. VandneguLu Saar
 a) Very good b) Thank you sir
 c) May I come Sir d) Good morning Sir
49. baalehaNNu
 a) Mango b) Chicku c) Banana d) Orange
50. ToTa _____
 a) Room b) Bangle c) House d) Garden

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Question Paper Version : B

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate the English question to Kannada :

1. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
2. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
3. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla
4. When are you going to College?
a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
5. Had your breakfast?
a) YaaVagu Tindi? b) TinDi ayitaa?
c) oota ayta? d) TinDi yestu beku?

Note : Write the English word for the given Kannada word :

6. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
7. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
8. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep

9. huTTu : _____
a) Good b) die c) Birth d) bed
10. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
11. daKShiNa : _____
a) West b) East c) South d) North
12. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
13. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
14. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
15. granthalaya : _____
a) House b) office c) college d) Library
16. baNNa : _____
a) White b) Banana c) Colour d) Red
17. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
18. pramaaNa : _____
a) promise b) very good c) purpse d) Thankyou
19. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
20. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
21. biLi : _____
a) red b) Black c) White d) Green
22. doDDa : _____
a) Big b) Small c) fat d) thin
23. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
24. gurugaLu
a) Former b) Teacher c) Father d) Uncle
25. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast

26. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give

27. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair

Note : Substitute the word from the following each sentence in appropriate places :

28. idu : idakke then adu _____
a) adakke b) idakke c) Horakke d) Horakkee

29. UURu : Urige then avaru _____
a) ivarige b) avarige c) Nimage d) Nawage

30. Mane : maneyalli then Pustaka
a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli

31. Raja : Rajanaddu, then Raani _____
a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Translate the Question Kannada to English :

32. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?

33. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none

34. ethaKKe hogidde?
a) Why did you go? b) Where did you go?
c) When you are going? d) None of these

35. nimma Kalejinaa hesarenu?
a) What is your college name? b) Where is your college
c) How old is your college? d) Which college are you studying?

36. naanu yenu MaDabhahudu?
a) What did you do? b) What should we do?
c) What can I do? d) None

37. eegale baa
a) come immediatly b) Go immediatly c) come Tomorrow d) None

38. dayamaDi niiVu oLagaDe banni
a) When you come b) Come inside
c) Don't come hear d) you please come inside

39. Kallegige raje KoTTi ddare
a) Holiday is given to college b) Don't come to college
c) Today college is there d) None

40. Raamanu nanaginta doDDavanu
 a) Rama is older than me
 c) Rama is my friend
 b) Rama is elder than me
 d) None
41. VandneguLu Saar
 a) Very good
 c) May I come Sir
 b) Thank you sir
 d) Good morning Sir
42. baalehaNNu
 a) Mango
 b) Chicku
 c) Banana
 d) Orange
43. ToTa _____
 a) Room
 b) Bangle
 c) House
 d) Garden

Fill in the blank by translating the given English word to Kannada :

44. You : _____
 a) neenu
 b) Naanu
 c) yaaru
 d) Ivaru
45. His : _____
 a) NeeVu
 b) avaru
 c) aVaLu
 d) avana
46. What : _____
 a) idu
 b) adu
 c) yenu
 d) yelli
47. Friend : _____
 a) Gumpu
 b) Maava
 c) Sneheta
 d) GurugaLu
48. Drama : _____
 a) naataka
 b) Kate
 c) haaDu
 d) Yaavadu alla
49. Choose the Mathematical + Symbol in Kannada
 a) Kale
 b) Koodu
 c) Jodisu
 d) Bhagisu
50. Country in Kannada will be called as
 a) D'washa
 b) Vesha
 c) Desha
 d) Vishesha

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Question Paper Version : C

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
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4. Darkening two circles for the same question makes the answer invalid.
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Note : Write the English word for the given Kannada word :

1. HaNNu : _____
a) seeds b) flower c) Tree d) Fruit
2. pramaaNu : _____
a) promise b) very good c) purpse d) Thankyou
3. tangi : _____
a) Aunti b) Uncle c) Sister d) Mother
4. gaNita : _____
a) Biology b) Physics c) English d) Mathematics
5. biLi : _____
a) red b) Black c) White d) Green
6. doDDa : _____
a) Big b) Small c) fat d) thin
7. Aarogya : _____
a) Tomarrow b) Health c) wealth d) cold
8. gurugaLu
a) Former b) Teacher c) Father d) Uncle
9. uuTa : _____
a) Meals b) Drinks c) Get up d) breakfast

10. KoDabeDa : _____
a) Give me b) Give us c) give d) Don't give
11. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
12. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
13. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
14. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
15. huTTu : _____
a) Good b) die c) Birth d) bed
16. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
17. daKShiNa : _____
a) West b) East c) South d) North
18. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
19. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
20. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
21. granthalaya : _____
a) House b) office c) college d) Library
22. baNNa : _____
a) White b) Banana c) Colour d) Red

Note : Translate the Question Kannada to English :

23. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?
24. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none
25. ethaKKe hogidde?
a) Why did you go? b) Where did you go?
c) When you are going? d) None of these

26. nimma Kalejinaa hesarenu?
 a) What is your college name? b) Where is your college
 c) How old is your college? d) Which college are you studying?
27. naanu yenu MaDabhahudu?
 a) What did you do? b) What should we do?
 c) What can I do? d) None
28. eegale baa
 a) come immediatly b) Go immediatly c) come Tomorrow d) None
29. dayamaDi niiVu oLagaDe banni
 a) When you come b) Come inside
 c) Don't come hear d) you please come inside
30. Kallegige raje KoTTi ddare
 a) Holiday is given to college b) Don't come to college
 c) Today college is there d) None
31. Raamanu nanaginta doDDavanu
 a) Rama is older than me b) Rama is elder than me
 c) Rama is my friend d) None
32. VandneguLu Saar
 a) Very good b) Thank you sir
 c) May I come Sir d) Good morning Sir
33. baalehaNNu
 a) Mango b) Chicku c) Banana d) Orange
34. ToTa _____
 a) Room b) Bangle c) House d) Garden

Fill in the blank by translating the given English word to Kannada :

35. You : _____
 a) neenu b) Naanu c) yaaru d) Ivaru
36. His : _____
 a) NeeVu b) avaru c) aVaLu d) avana
37. What : _____
 a) idu b) adu c) yenu d) yelli
38. Friend : _____
 a) Gumpu b) Maava c) Sneheta d) GurugaLu
39. Drama : _____
 a) naataka b) Kate c) haaDu d) Yaavadu alla

40. Choose the Mathematical + Symbol in Kannada
 a) Kale b) Koodu c) Jodisu d) Bhagisu
41. Country in Kannada will be called as
 a) D'weshu b) Vesha c) Desha d) Vishesha

Note : Translate the English question to Kannada :

42. Where is he?
 a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
43. Where shall we meet?
 a) NaVu yaaru? b) Sanje barattiya?
 c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
44. When is the physics class?
 a) physics class yaavagaa? b) Sanje physicedeyan?
 c) physic Lab elli ede? d) yaavadu alla
45. When are you going to College?
 a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
 c) neenu yavagu hogutitya? d) enu MaDuttiddiya?
46. Had your breakfast?
 a) YaaVagu Tindi? b) TinDi ayitaa?
 c) oota ayta? d) TinDi yestu beku?

Note : Substitute the word from the following each sentence in appropriate places :

47. idu : idakke then adu _____
 a) adakke b) idakke c) Horakke d) Horakkee
48. UUru : Urige then avaru _____
 a) ivarige b) avarige c) Nimage d) Nawage
49. Mane : maneyalli then Pustaka
 a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
50. Raja : Rajanaddu, then Raani _____
 a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

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Question Paper Version : D

Third Semester B.E. /B.Tech. Degree Examination, Jan./Feb. 2023
Balake Kannada

[Time: 1 hrs.]

[Max. Marks: 50]

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the Fifty questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

Note : Translate the Question Kannada to English :

1. eshTu ganTe Saar?
a) Can I go Sir? b) will I come Sir?
c) 8 Clock Sir d) What is the time Sir?
2. Ee pustaKa yaaraduu?
a) Is that book yours b) whose book is this?
c) This book belongs to me. d) none
3. ethaKKe hogidde?
a) Why did you go? b) Where did you go?
c) When you are going? d) None of these
4. nimma Kalejinaa hesarenu?
a) What is your college name? b) Where is your college
c) How old is your college? d) Which college are you studying?
5. naanu yenu MaDabhahudu?
a) What did you do? b) What should we do?
c) What can I do? d) None
6. eegale baa
a) come immediatly b) Go immediatly c) come Tomorrow d) None
7. dayamaDi niiVu oLagaDe banni
a) When you come b) Come inside
c) Don't come hear d) you please come inside

8. Kallegige raje KoTTi ddare
a) Holiday is given to college b) Don't come to college
c) Today college is there d) None
9. Raamanu nanaginta doDDavanu
a) Rama is older than me b) Rama is elder than me
c) Rama is my friend d) None
10. VandneguLu Saar
a) Very good b) Thank you sir
c) May I come Sir d) Good morning Sir
11. baalehaNNu
a) Mango b) Chicku c) Banana d) Orange
12. ToTa _____
a) Room b) Bangle c) House d) Garden

Fill in the blank by translating the given English word to Kannada :

13. You : _____
a) neenu b) Naanu c) yaaru d) Ivaru
14. His : _____
a) NeeVu b) avaru c) aVaLu d) avana
15. What : _____
a) idu b) adu c) yenu d) yelli
16. Friend : _____
a) Gumpu b) Maava c) Sneheta d) GurugaLu
17. Drama : _____
a) naataka b) Kate c) haaDu d) Yaavadu alla
18. Choose the Mathematical + Symbol in Kannada
a) Kale b) Koodu c) Jodisu d) Bhagisu
19. Country in Kannada will be called as
a) D'wsha b) Vesha c) Desha d) Vishesha

Note : Translate the English question to Kannada :

20. Where is he?
a) avanu elliddane? b) nanu yellige bande? c) AvaLu yaaru? d) idu yaavadu?
21. Where shall we meet?
a) NaVu yaaru? b) Sanje barattiya?
c) enu MaDuttiddiya? d) naavu elli bheTi aagoNO?
22. When is the physics class?
a) physics class yaavagaa? b) Sanje physicedeyan?
c) physic Lab elli ede? d) yaavadu alla

23. When are you going to College?
 a) Neenu yaavaaga Kalejige hoguttiya? b) Yalli huguttiya?
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Note : Substitute the word from the following each sentence in appropriate places :

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 a) Pustakadalli b) peuninalli c) rajeyalli d) nammalli
28. Raja : Rajanaddu, then Raani _____
 a) Raviyaddu b) Ruehiyaddu c) Raniyaddu d) Hudugiyaddu

Note : Write the English word for the given Kannada word :

29. HaNNu : _____
 a) seeds b) flower c) Tree d) Fruit
30. pramaaNu : _____
 a) promise b) very good c) purpse d) Thankyou
31. tangi : _____
 a) Aunti b) Uncle c) Sister d) Mother
32. gaNita : _____
 a) Biology b) Physics c) English d) Mathematics
33. biLi : _____
 a) red b) Black c) White d) Green
34. doDDa : _____
 a) Big b) Small c) fat d) thin
35. Aarogya : _____
 a) Tomarrow b) Health c) wealth d) cold
36. gurugaLu
 a) Former b) Teacher c) Father d) Uncle
37. uuTa : _____
 a) Meals b) Drinks c) Get up d) breakfast
38. KoDabeDa : _____
 a) Give me b) Give us c) give d) Don't give

39. Kaagada : _____
a) Mouse b) Computer c) Paper d) Chair
40. Aangla : _____
a) Kannada b) English c) Telugu d) Tamil
41. Vidyarthi : _____
a) Student b) Teacher c) Principal d) HOD
42. SamSkruti : _____
a) Culture b) festival c) Haritage d) Weep
43. huTTu : _____
a) Good b) die c) Birth d) bed
44. aalugeDDe : _____
a) fruits b) Cabbage c) Potato d) Tomato
45. daKShiNa : _____
a) West b) East c) South d) North
46. Navilu : _____
a) Hen b) Peacock c) Ant d) Parrot
47. MaNNu : _____
a) Soil b) Stone c) Air d) Hill
48. HeNDati : _____
a) Sun b) aKKa c) Wife d) sister
49. granthalaya : _____
a) House b) office c) college d) Library
50. baNNa : _____
a) White b) Banana c) Colour d) Red
